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Section: B

LAB REPORT NO.2

Exercises: LAB 02

TASK 1:

Write a class Student with three data members: RollNo, Name, and Marks. The class should have:

- A default constructor to initialize values to 0 or empty.
- A parameterized constructor to initialize values using arguments.
- A display() function to print the details.
- In main(), create two objects: one with default values and another with provided values.

SOURCE CODE:

```
#include <iostream>
using namespace std;
```

```
class Student {
public:
```

```
    int RollNo;
    string Name;
    float Marks;
```

```
    Student() {
        RollNo = 0;
        Name = "";
        Marks = 0;
    }
```

```
    Student(int r, string n, float m) {
        RollNo = r;
        Name = n;
        Marks = m;
    }
```

```
    void display() {
        cout << "RollNo: " << RollNo << endl;
        cout << "Name: " << Name << endl;
```

```
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  public:
6      int RollNo;
7      string Name;
8      float Marks;
9
10     Student() {
11         RollNo = 0;
12         Name = "";
13         Marks = 0;
14     }
15
16     Student(int r, string n, float m) {
17         RollNo = r;
18         Name = n;
19         Marks = m;
20     }
21
22     void display() {
23         cout << "RollNo: " << RollNo << endl;
24         cout << "Name: " << Name << endl;
25         cout << "Marks: " << Marks << endl;
26     }
27 };
28
29 int main() {
30     Student s1;
31     Student s2(1, "Ali", 85.5);
32
33     s1.display();
34     s2.display();
35
36     return 0;
37 }
```

```

        cout << "Marks: " << Marks << endl;
    }
};

int main() {
    Student s1;
    Student s2(1, "Ali", 85.5);

    s1.display();
    s2.display();

    return 0;
}

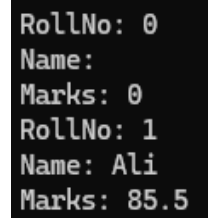
```

OUTPUT:

```

RollNo: 0
Name:
Marks: 0
RollNo: 1
Name: Ali
Marks: 85.5

```



```

RollNo: 0
Name:
Marks: 0
RollNo: 1
Name: Ali
Marks: 85.5

```

TASK 2:

Write a class Rectangle with two private data members: length and width. The class should have:

- i. A parameterized constructor to initialize values.
- ii. A copy constructor that initializes a new object with values from another object.
- iii. An area() function to return the area.
- iv. In main(), create an object and then create a second object using the copy constructor, displaying both areas.

SOURCE CODE:

```

#include <iostream>
using namespace std;

class Rectangle {
private:
    int length;
    int width;

public:
    Rectangle(int l, int w) {
        length = l;
        width = w;
    }

    Rectangle(const Rectangle &r) {

```

```

        length = r.length;
        width = r.width;
    }

    int area() {
        return length * width;
    }
};

int main() {
    Rectangle r1(5, 4);
    Rectangle r2 = r1;

    cout << "Area of r1: " << r1.area() << endl;
    cout << "Area of r2: " << r2.area() << endl;

    return 0;
}

```

```

1  #include <iostream>
2  using namespace std;
3
4  class Rectangle {
5  private:
6      int length;
7      int width;
8
9  public:
10     Rectangle(int l, int w) {
11         length = l;
12         width = w;
13     }
14
15     Rectangle(const Rectangle &r) {
16         length = r.length;
17         width = r.width;
18     }
19
20     int area() {
21         return length * width;
22     }
23 };
24
25 int main() {
26     Rectangle r1(5, 4);
27     Rectangle r2 = r1;
28
29     cout << "Area of r1: " << r1.area() << endl;
30     cout << "Area of r2: " << r2.area() << endl;
31
32     return 0;
33 }

```

OUTPUT:

```

Area of r1: 20
Area of r2: 20

```

```

Area of r1: 20
Area of r2: 20

```

TASK 3:

Write a class Car with three private data members: Model, Company, and Price. The class should have:

- i. A parameterized constructor to initialize values.
- ii. A destructor that prints "Object Destroyed" when an object is deleted.
- iii. A display() function to print details.
- iv. In main(), create two objects and observe when the destructor is called.

SOURCE CODE:

```

#include <iostream>
using namespace std;

class Car {
private:
    string Model;
    string Company;
    float Price;

public:
    Car(string m, string c, float p) {
        Model = m;
        Company = c;
        Price = p;
    }
}

```

```

~Car() {
    cout << "Object Destroyed" << endl;
}

void display() {
    cout << "Model: " << Model << endl;
    cout << "Company: " << Company << endl;
    cout << "Price: " << Price << endl;
}

};

int main() {
    Car c1("Civic", "Honda", 30000);
    Car c2("Corolla", "Toyota", 25000);

    c1.display();
    c2.display();

    return 0;
}

```

OUTPUT:

```

Model: Civic
Company: Honda
Price: 30000
Model: Corolla
Company: Toyota
Price: 25000
Object Destroyed
Object Destroyed

```

```

1  #include <iostream>
2  using namespace std;
3
4  class Car {
5  private:
6      string Model;
7      string Company;
8      float Price;
9
10 public:
11     Car(string m, string c, float p) {
12         Model = m;
13         Company = c;
14         Price = p;
15     }
16
17     ~Car() {
18         cout << "Object Destroyed" << endl;
19     }
20
21     void display() {
22         cout << "Model: " << Model << endl;
23         cout << "Company: " << Company << endl;
24         cout << "Price: " << Price << endl;
25     }
26 };
27
28 int main() {
29     Car c1("Civic", "Honda", 30000);
30     Car c2("Corolla", "Toyota", 25000);
31
32     c1.display();
33     c2.display();
34
35     return 0;
36 }

```

```

Model: Civic
Company: Honda
Price: 30000
Model: Corolla
Company: Toyota
Price: 25000
Object Destroyed
Object Destroyed

```

TASK 4:

Write a class Circle with one private data member: radius. The class should have:

- i. A default constructor that initializes radius = 1.
- ii. A parameterized constructor that sets radius based on an argument.
- iii. A circumference() function that returns the circumference.
- iv. In main(), create two objects using both constructors and display their circumferences.

SOURCE CODE:

```

#include <iostream>
using namespace std;

class Circle {
private:
    double radius;

```

```
public:
    Circle() {
        radius = 1;
    }

    Circle(double r) {
        radius = r;
    }

    double circumference() {
        return 2 * 3.14 * radius;
    }
};

int main() {
    Circle c1;
    Circle c2(5);

    cout << "Circumference of c1: " << c1.circumference() << endl;
    cout << "Circumference of c2: " << c2.circumference() << endl;

    return 0;
}
```



```
1  #include <iostream>
2  using namespace std;
3
4  class Circle {
5  private:
6      double radius;
7
8  public:
9      Circle() {
10         radius = 1;
11     }
12
13     Circle(double r) {
14         radius = r;
15     }
16
17     double circumference() {
18         return 2 * 3.14 * radius;
19     }
20 };
21
22 int main() {
23     Circle c1;
24     Circle c2(5);
25
26     cout << "Circumference of c1: " << c1.circumference() << endl;
27     cout << "Circumference of c2: " << c2.circumference() << endl;
28
29     return 0;
30 }
```

OUTPUT:

Circumference of c1: 6.28
Circumference of c2: 31.4

```
Circumference of c1: 6.28
Circumference of c2: 31.4
```

TASK 5:

Write a class Resource with a private data member value. The class should have:

- i. A parameterized constructor to initialize value.
- ii. A move constructor to transfer ownership.
- iii. A display() function to print the value.
- iv. In main(), create an object and then use move() to initialize another object, displaying both values.

SOURCE CODE:

```
#include <iostream>
using namespace std;

class Resource {
private:
    int value;

public:
    Resource(int v) {
        value = v;
    }

    Resource(const Resource &r) {
        value = r.value;
    }

    void display() {
        cout << "Value: " << value << endl;
    }
};

int main() {
    Resource r1(100);
    Resource r2 = r1;

    r1.display();
    r2.display();

    return 0;
}
```

```
1  #include <iostream>
2  using namespace std;
3
4  class Resource {
5  private:
6      int value;
7
8  public:
9      Resource(int v) {
10         value = v;
11     }
12
13     Resource(const Resource &r) {
14         value = r.value;
15     }
16
17     void display() {
18         cout << "Value: " << value << endl;
19     }
20 };
21
22 int main() {
23     Resource r1(100);
24     Resource r2 = r1;
25
26     r1.display();
27     r2.display();
28
29     return 0;
30 }
```

OUTPUT:

Value: 100
Value: 100

```
Value: 100
Value: 100
```